2022 exam - Bayesian

Thursday, 27 October 2022 2:01 pm

$x \sim N(\mu, 1^2)$

$y \sim N(0, \frac{1}{\tau})$

$p(\mu, \tau) = p(\mu|\tau)p(\tau)$

$\mu|\tau \sim N(0, \frac{1}{\tau})$ $\quad p(\mu|\tau) = \sqrt{\frac{\tau}{2\pi}}\, e^{-\frac{\tau\mu^2}{2}}$

$\tau \sim \text{Gamma}(a=2, b=1)$

$p(\tau) = \frac{1}{\Gamma(2)}\tau e^{-\tau}$

(a)

$p(x,y|\mu,\tau)\,p(\mu,\tau)$

$\propto e^{-\frac{(x-\mu)^2}{2}} \cdot \sqrt{\frac{\tau}{2\pi}}\, e^{-\frac{\tau y^2}{2}} \cdot \sqrt{\frac{\tau}{2\pi}}\, e^{-\frac{\tau\mu^2}{2}} \cdot \tau e^{-\tau}$

$p(\mu|\tau) \propto p(x,y|\mu,\tau)\,p(\mu,\tau)$

$\propto e^{-\frac{(1+\tau)\mu^2 - 2x\mu}{2}}$

$\sim N\left(\frac{x}{1+\tau}, \frac{1}{1+\tau}\right)$ – (a)

~~$p(\tau|\mu) \propto p(x,y|\mu,\tau)\,p(\mu,\tau)$~~

~~$\propto \tau^2 e^{-\tau\left(\frac{y^2+\mu^2}{2}+1\right)}$~~

~~$\sim \text{Gamma}\left(3, \frac{y^2+\mu^2}{2}+1\right)$~~

(c)

~~$\log q^*_\mu(\mu) = E_\tau\left[\log p(x,y|\mu,\tau)\,p(\mu,\tau)\right] + \text{const}$~~

~~$= E_\tau\left[-\frac{(x-\mu)^2}{2} + 2\log\tau - \frac{\tau}{2}y^2 - \frac{\tau}{2}\mu^2\right] + \text{const}$~~

~~$= -\frac{(x-\mu)^2}{2} - \frac{\mu^2}{2}E_\tau(\tau) + \text{const}$~~

~~$= -\frac{[1+E_\tau(\tau)]\mu^2 - 2x\mu}{2} + \text{const}$~~

~~pdf of $N(\mu^*, \sigma^{2*})$~~

~~$\mu^* = \frac{x}{1+E_\tau(\tau)}$ $\quad \sigma^{2*} = \frac{1}{1+E_\tau(\tau)}$~~

$\log q^*_\tau(\tau) = E_\mu\left[\log p(x,y|\mu,\tau)\,p(\mu,\tau)\right] + \text{const}$

$= E_\mu\left[-\frac{(x-\mu)^2}{2} + 2\log\tau - \frac{\tau}{2}y^2 - \frac{\tau}{2}\mu^2 - \tau\right] + \text{const}$

$= 2\log\tau - \frac{\tau}{2}y^2 - \frac{\tau}{2}E_\mu(\mu^2) - \tau + \text{const}$

$= 2\log\tau - \tau\left(\frac{y^2 + E_\mu(\mu^2)}{2} + 1\right)$

Gamma(a, b)

↑ shape ↑ scale

$a = 3 \quad b = \frac{2}{\left(y^2 + \underbrace{E_\mu(\mu^2)}_{V_\mu(\mu) + (E_\mu(\mu))^2} + 2\right)}$

(c)

$\tau_n \sim \exp(\text{rate} = \frac{1}{\tau_c}) \Rightarrow \frac{1}{\tau_c} e^{-\frac{\tau_n}{\tau_c}}$

$\mu_n \sim N(\text{mean} = 0, \text{variance} = \frac{1}{\tau_n}) \Rightarrow \sqrt{\frac{\tau_n}{2\pi}}\, e^{-\frac{\tau_n\mu_n^2}{2}}$

(b)

$x = 0, \quad y = 1 \quad \theta_c = (\mu_c, \tau_c) = (-1, 2)$

$\theta_n = (\mu_n, \tau_n) = (1, 1)$

$p(x,y|\mu,\tau)\,p(\mu,\tau)$

$\propto e^{-\frac{(x-\mu)^2}{2}} \cdot \sqrt{\frac{\tau}{2\pi}}\, e^{-\frac{\tau y^2}{2}} \cdot \sqrt{\frac{\tau}{2\pi}}\, e^{-\frac{\tau\mu^2}{2}} \cdot \tau e^{-\tau} \propto \tau^2 e^{-\frac{1}{2}\left((x-\mu)^2 + \tau y^2 + \tau\mu^2 + 2\tau\right)}$

$AP = \min\left\{1, \boxed{\frac{p(x,y|\theta^n)\,p(\theta^n)}{p(x,y|\theta^c)\,p(\theta^c)}}^{☆} \boxed{\frac{p(\theta^c|\theta^n)}{p(\theta^n|\theta^c)}}^{*}\right\}$

$☆ = \frac{1\, e^{-\frac{1}{2}(1^2 + 1 + 1 + 2)}}{4\, e^{-\frac{1}{2}(1^2 + 2 + 2 + 4)}} = \frac{1}{4}e^{2}$

$* = \frac{\frac{1}{\tau_n} e^{-\frac{\tau_c}{\tau_n}} \sqrt{\frac{\tau_c}{\cancel{2\pi}}}\, e^{-\frac{\tau_c\mu_c^2}{2}}}{\frac{1}{\tau_c} e^{-\frac{\tau_n}{\tau_c}} \sqrt{\frac{\tau_n}{\cancel{2\pi}}}\, e^{-\frac{\tau_n\mu_n^2}{2}}}$

$= \frac{\frac{1}{1} e^{-\frac{2}{1}} \sqrt{2}\, e^{-\frac{2\cdot 1}{2}}}{\frac{1}{2} e^{-\frac{1}{2}} \sqrt{1}\, e^{-\frac{1}{2}}} = 2\sqrt{2}\, e^{-2\cancel{-1}\cancel{+1}}$

$AP = \min\left\{1, \frac{\sqrt{2}}{2}\right\} = \boxed{0.7071068}$

$= 0.7071068$

(b)